



## EDUCATION

| Year | Degree/Exam                                     | Institute                            | CGPA/Marks |
|------|-------------------------------------------------|--------------------------------------|------------|
| 2021 | M.TECH Dual Degree 5Y                           | IIT Kharagpur                        | 8.72 / 10  |
| 2016 | All India Senior School Certificate Examination | Central Board of Secondary Education | 93.8%      |
| 2014 | All India Secondary School Examination          | Central Board of Secondary Education | 10 / 10    |

## SKILLS AND EXPERTISE

- **Languages:** C++, C, Python, Matlab
- **Libraries and Tools:** C++ STL, PyTorch, Numpy, OpenCV, SciPy, Scikit-learn, Pandas
- **Fields of Interest:** Design and Analysis of Algorithms, Deep Learning, Machine Learning, Computer Vision

## INTERSHIPS

**Research Internship at Insight Center for Data Analytics NUI Galway (remote, June 2020 - August 2020)***Clustering High Dimensional Vector Space for Efficient Peer-to-Peer Network Routing*

- Worked on Clustering High Dimensional Vector Space for Efficient Peer-to-Peer Network Routing under the supervision of Dr. Edward Curry.
- Designed and implemented semantic clustering algorithms on high-dimensional vector space for efficient routing in decentralised peer-to-peer networks dealing with unbounded content spaces.

**Research Internship at University of Alberta (University of Alberta Research Experience programme, May 2019 - July 2019)***3D Ultrasound Data Segmentation using Deep Learning*

- Worked on Segmentation of Subcortical Structures of 3D Brain MRI Data using Deep Learning, under the supervision of Dr. Dana Cobzas, Assistant Adjunct Professor, Department of Computing Science, University of Alberta, in collaboration with Peter S. Allen MR Research Centre.
- Implemented a deep learning based automated system for segmenting subcortical brain structures, for use by the Peter S. Allen MR Research Centre for research and other purposes.

*Semi-supervised Segmentation of Medical Images using Deep Learning*

- Worked on a semi-supervised method of segmenting medical images, under the supervision of Dr. Dana Cobzas.
- Designed a novel deep neural architecture by incorporating random walker model into the segmentation network.

## PROJECTS

**Mutual Translation between CT and Cone-Beam CT of Human Lungs (B.Tech Project, Spring 2020)**

- Worked on Mutual Translation between Computed Tomography and Cone-Beam Computed Tomography of Human Lungs as part of B.Tech Project, under the supervision of Dr. Jayanta Mukhopadhyay, Professor, Department of Computer Science and Engineering, IIT Kharagpur.
- Implemented a Cycle-GAN based deep learning model for mutual translation between CT and CBCT images of human lungs.

**Application of Soft Computing in Selective Grooming and Monitoring of Students (Spring 2019)**

- Designed machine learning models, neural networks and fuzzy inference systems to predict a student's academic performance based on lifestyle and past academic records, for the purpose of Selective Grooming and Monitoring of Students, as a term project for CE31501: Soft Computing Tools in Engineering.

**Traffic Sign Recognition using Image Processing (Autumn 2018)**

- Worked on Traffic Sign Recognition using Image Processing Techniques as a term project for the course CS40019: Image Processing.

**Segmentation of Nuclei in Pathological Images using Neural Networks (May 2018 - July 2018)**

- Worked on Segmentation of Nuclei in Pathological Images using Neural Networks at Signal and Image Processing Lab, IIT Kharagpur, under the supervision of Dr. Debdeep Sheet, Department of Electrical Engineering, IIT Kharagpur.

## COURSEWORK INFORMATION

**Academic:** Programming and Data Structures\*, Algorithms-I\*, Transform Calculus, Image Processing, Digital Electronics\*, Control Systems\*, Soft Computing Tools in Engineering, Computer Networks, Computer Architecture and Operating System, Embedded Systems\*, Probability and Stochastic Processes, Digital Signal Processing, Machine Learning, Statistical Signal Processing, Advanced Digital Image Processing and Computer Vision.

**Self-Learning:** Coursera certified online courses (Stanford University): Greedy Algorithms, Minimum Spanning Trees, and Dynamic Programming, Graph Search, Shortest Paths, and Data Structures, Divide and Conquer, Sorting and Searching, and Randomized Algorithms; Deep Learning MOOC offered by Andrew Ng in Coursera; Coursera certified online courses (University of Michigan): Statistics with Python Specialisation (Understanding and Visualizing Data with Python, Inferential Statistical Analysis with Python, Fitting Statistical Models to Data with Python), Introduction to Data Science in Python, Applied Plotting, Charting & Data Representation in Python.

*Courses marked by a '\*' also have a laboratory component.*

## POSITIONS OF RESPONSIBILITY

**Mentor at Student Welfare Group, IIT Kharagpur (July 2018 - present)**

Involved in mentoring junior students as a Mentor at Student Welfare Group, IIT Kharagpur under its mentorship programme.

**Teaching Assistant for Control and Instrumentation Lab - EE39009, IIT Kharagpur (Autumn 2020)**

Assisted in the conduction of Control and Instrumentation Lab (Autumn 2020) in the department of Electrical Engineering.

## EXTRA CURRICULAR ACTIVITIES

- Involved in various social service activities aimed at the welfare of the society, especially in rural areas, as a student volunteer of National Service Scheme (NSS), a Central Sector Scheme of Government of India, Ministry of Youth Affairs and Sports.
- Part of Silver medal winning unit in NSS Annual Camp 2016, IIT Kharagpur, organised at Khelari, West Midnapore, West Bengal, India.